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The Developmental English Lexicon Project (d-ELP): Continuous Word Reading Difficulty

Estimates of the 9,961 Most Frequently Printed English Words for US Children

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The d-ELP relational database can be accessed online at: <http://d-elp.org>

Supplemental materials can be accessed online at:

https://osf.io/ar9hv/overview?view_only=323ee6d43e204f4781a9802abfbcdf71

Abstract

The developmental English Lexicon Project (d-ELP) is a database comprising word reading data for 9,961 of the most frequently printed English words for children, including continuous IRT-based estimates of word reading difficulty, accompanying word characteristics, and raw trial-level data. Data on isolated word reading accuracy were collected from 1,907 children in grades 1-5 across multiple US sites. Each child read a list of words ranging from 250 to 500 items depending on reading skill (less proficient children were often unable to finish the entire list of words), with each list representing the full distribution of word difficulties oversampled for words at the child's reading level. Each child also attempted 99 linking items representing the full distribution of word difficulties. A total of 631,322 total responses were collected with the average child providing 336. For the final sample of words, the average number of responses per word for the 9,862 non-linking items was 47.14 (range of 10-592) and for the 99 linking items the average was 1681.42 (range of 1083-1907). A Rasch model, with planned missing design, was fit to children's responses on the word list and linking item responses. This introduction describes the motivation for this project; the methods used for data collecting; the analytic framework used to establish a continuous measure of word reading difficulty across words; descriptive statistics of the relational database and initial behavioral results; and a publicly available search engine providing searchable access to the word difficulty ratings and associated word-level statistics. Abstract Word Count: 250

The Developmental English Lexicon Project (d-ELP): Continuous Word Reading Difficulty

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The English orthography exhibits probabilistic, rather than strictly rule-governed, mappings between orthography and phonology across multiple grain sizes (see Ziegler, & Goswami, 2005; Venezky, 1999). Its structure reflects a complex interplay of morphological regularities, syllabic constraints, and numerous sublexical regularities in spelling–sound correspondences – features that have led Seidenberg (2005, 2017) and colleagues (e.g., Plaut et al., 1996; Rueckl & Seidenberg, 2011; Seidenberg et al., 2020; Seidenberg & McClelland, 1989) to characterize English as *quasiregular*. For this reason, English poses unique and significant challenges for children as they work to develop efficient and automatic word reading skills (see Seymour, 2013; Seymour et al., 2003). Consequently, the English orthography gives rise to substantial individual differences in word reading accuracy and speed in developing readers, reflecting the joint effects of various word-level characteristics such as frequency, length, and orthography-to-phonology and orthography-to-semantics consistency (for details, see Castles, Rastle, & Nation, 2018; Compton et al., 2014; Deacon & Levesque, 2024; Kearns, 2015, 2020; Kearns et al., 2016; Kellenberger et al., 2024; Leslie & Calhoun, 1995; Martens & de Jong, 2006; Marinus, Nation, & de Jong, 2015; Rueckl, 2016; Seidenberg, 1989; Siegelman et al., 2020; Steacy & Compton, 2019; Steacy et al., 2022; Treiman et al., 1990; Wade-Woolley & Heggie, 2015; Waters, Seidenberg, & Bruck, 1984).

Attempts to explore, understand, and predict the combined effects of multiple word-level characteristics on word reading accuracy in developing readers have been constrained by several limitations in the developmental reading literature. These include: (1) although the field has produced several large-scale databases of word recognition – commonly referred to as “lexicon

projects” and related “crowdsourcing projects” (e.g., Adelman et al., 2014; Balota et al., 2007; Keuleers et al., 2012; Mander et al., 2020) – these resources primarily reflect adult reading behavior (but see Schröter & Schroeder, 2017, for a developmental database of German word reading); (2) empirical studies of word reading acquisition have predominantly focused on monosyllabic, monomorphemic words (e.g., Balota et al., 2007; Coltheart et al., 2001; Perry et al., 2010; Roberts et al., 2011; Seidenberg & McClelland, 1989; Siegelman et al., 2020; Yap & Balota, 2009); (3) most developmental studies rely on small samples of both children and words and employ factorial designs with limited control variables (e.g., Strain, Patterson, & Seidenberg, 2002); and (4) the absence of word difficulty estimates in publicly available databases that report lexical and functional orthographic characteristics for words commonly found in children’s print input, including both published (e.g., Korochkina & Rastle, 2025) and unpublished (e.g., Dawson et al., 2023) sources.

To help alleviate these important limitations within the developmental word reading literature we introduce the developmental English lexicon project (d-ELP), a publicly available relational database containing word reading difficulty ratings, based on child single word recognition/reading performance, for a large number of words found in print. The d-ELP comprises continuous IRT-based word reading difficulty estimates for 9,961 of the most frequently printed English words for US children in grades 1-5, along with accompanying word characteristics. The database was designed to allow detailed model testing of word characteristics affecting word reading development across a large number of words in a relatively diverse sample of US English-speaking children. Other possible uses of the d-ELP include the validation of newly developed and age-appropriate psycholinguistic variables, sub-sampling of specific items to conduct “virtual experiments” (see, e.g., Kuperman, 2015), and item selection for

developmental, experimental, and individual differences studies. The database provides developmental data to support the evaluation and advancement of computational models of word reading development, raw materials for creating instructional materials that are appropriate for developing readers and spellers, including AI-based assessment and instructional programs, and precise data to allow the development of age-appropriate and free-of-bias summative and formative assessments of word reading skills. This work was further motivated by recent questions regarding the magnitude of impacts that various word-level variables have on word reading performance and which variables are required as control variables in factorial experiments and individual differences studies (for details see, Balota, Yap, Hutchison, & Cortese, 2012; Brysbaert, Keuleers, & Mander, 2014; Keuleers & Balota, 2015).

In this initial introduction to the d-ELP database we detail the motivation for the project, the methods used to collect the data, the analytic framework used to establish a continuous measure of word reading difficulty across a wide swath of printed words, the descriptive statistics of the database, and an overview of the functionality of the publicly available relational database that provides searchable access to the word difficulty ratings and associated word-level statistics for each of the stimuli (along with the raw trial-level data). We also demonstrate the d-ELP's ability to estimate the impact of general word-level predictors (e.g., frequency, word length, transparency, semantic measures, and orthographic neighborhood size) on word recognition performance in developing readers to help demonstrate the potential of the database to inform mechanisms associated with early word reading development and identify important control variables to be included in smaller scale experiments of word reading performance in developing readers.

Predicting English Word Difficulty in Developing Readers

Children's word reading development follows a curvilinear growth pattern, with more rapid progress in early childhood and slower growth as they get older (for details see Francis et al., 1996). By eighth grade, a typically developing reader can recognize roughly 10,000 unique word forms, excluding inflectional variants (Ehri, 2005; Harris & Jacobson, 1982). The development of word reading skill in children has traditionally been characterized at a micro level by item-learning processes (e.g., Share, 1995) and at a macro level by stage-based processing (Ehri, 2002). Item learning refers to the process by which children gradually establish high-quality lexical representations through phonological recoding and repeated encounters with print. Through the process of fully analyzing each word, typically through phonological recoding, a unique orthographic entry is added to the lexicon, enabling efficient, automatic retrieval and access to phonological, syntactic, and semantic information during reading (Castles et al., 2018; Humphreys, 1985; Perfetti, 1992; Perfetti & Hart, 2001; Perfetti & Stafura, 2014; Snow, 2002; Stanovich, 1991). Stage-based theories, in contrast, describe developmental regularities in the strategies and knowledge, both implicit and explicit, that children bring to word identification. These models specify how a child's available linguistic knowledge – phonological, orthographic, and semantic – conditions the probability of successfully reading a given word, thereby linking child-level skill with word-level properties. Early stage-based models characterized this evolution as a series of qualitative shifts from visual recognition to phonological decoding and ultimately to automatic word recognition (e.g., Chall, 1983; Ehri, 1995; Frith, 1985). Modern perspectives increasingly view these stages not as discrete, but as emergent from continuous item-level learning. Children's experience-driven tuning of orthography–phonology and orthography–semantics mappings lead to the apparent stage-like patterns observed at the behavioral level, while promoting expanded and strengthened

connections between orthography, phonology, and semantics (Siegelman, Kearns, & Rueckl, 2020). This evolving lexical organization enables children to generalize learned patterns to increasingly varied and challenging forms, ultimately allowing them to process novel and ever more complex letter strings with greater flexibility and efficiency (see Siegelman, Rueckl, et al., 2020).

Child-by-Word Models of Word Reading Development

The emergence of explanatory item response models (see De Boeck & Wilson, 2004) has allowed the merging of micro- and macro-perspectives into what is now referred to as “child-by-word” models of word reading development (e.g., Harm & Seidenberg, 2004; Nation & Snowling, 1988; Steacy et al., 2022). Child-by-word models of word reading development assert that: word reading is not uniform across words or readers; the same child may read different words with varying success depending on word properties; the same word may be read differently by different children depending on a child’s reading skills and experience; and thus, reading development involves interactions between the attributes of the word and characteristics of the child, including experiences with text (e.g., Kellenberger et al., 2024; Petscher et al., 2020; Steacy, 2020). These child-by-word differences are shaped, in large part, by children’s statistical learning of orthography–phonology and orthography-semantic correspondences (see Siegelman, Rueckl, et al., 2020). Through repeated exposure to print, statistical learning mechanisms allow children to detect and internalize the probabilistic relationships between written and spoken word forms, becoming increasingly sensitive to the covariance structure linking orthographic patterns with their phonological and semantic representations in English (e.g., Apfelbaum, Hazeltine, & McMurray, 2013; Chetail, 2015; Nation, 2017; Treiman & Kessler, 2006). This growing sensitivity supports the development of a probabilistic mapping system that facilitates decoding

and word recognition, enabling readers to navigate the quasiregular nature of the English orthography – capitalizing on high-frequency patterns while flexibly adapting to less common mappings (Sawi & Rueckl, 2019; Steacy et al., 2019; Treiman et al., 2003, 2006).

To illustrate, early connections between orthography and phonology in developing readers are initially based on simple grapheme-phoneme correspondences that are relatively insensitive to orthographic context (see de Jong & Molenaar, 2025; Share, 1995). With reading development, initially incomplete and oversimplified representations evolve to become more sophisticated context-dependent connections (see Steacy et al., 2019; Treiman et al., 2003, 2006). A good example of this is the *ea* diagraph which appears frequently in one-syllable words and is inconsistent in its pronunciation. As children develop word reading skill and practice reading in text, they encounter various pronunciations of *ea*, such as /i/ in *meat*, /ɛ/ in *head*, and /eɪ/ in *great* along with other less frequent pronunciations (e.g., /ə/ in *ocean* and /I/ in *Guinea*). Experiences with “*ea*-words” shape and influence the nature of orthography-to-phonological connection for *ea* slowly evolving into a set of representations that mirror the probabilistic relationship between orthography and phonology for *ea* in the English corpus. These sub-lexical connections between orthographic and phonology are influenced by the regularity/consistency of the diagraph pronunciation and the frequency of the words with which the diagraph appears. Figure 1 presents the relationship between word reading difficulty (based on the d-ELP database), spelling-to-pronunciation transparency (a proxy for word regularity), and word frequency for single syllable words with the spelling pattern *-eat*. One can see in the figure the well-established word frequency by regularity effect (see Seidenberg et al., 1984) in which the difficulty of high frequency words is not influenced by regularity (e.g., *eat* vs. *great*), whereas in low frequency words regularity of pronunciations affects difficulty (e.g., *cheat* vs. *sweat*).

Figure 1 Here

Evidence for the child-by-word perspective can be further illustrated by examining the relationship between children’s word reading skill and their reading performance on “*ea*-nonwords” (see Steacy et al., 2019; Treiman et al., 2003, 2006). In the Steacy study, the probability of reading a nonword such as *zeam* vs. *zead* (with either the higher frequency pronunciation of *ea* as /i/ vs. the lower frequency pronunciation of *ea* as /ɛ/) was dependent on a child’s word reading skill and the corpus-based rime support for the vowel pronunciation (*-eam* supports *ea* as /i/ whereas *-ead* supports *ea* as /ɛ/). Results suggest that general reading skill at the child level and rime support at the nonword level uniquely facilitate context-dependent vowel pronunciations in our sample of developing readers (i.e., *zeam* read to rhyme with *team* and *zead* to rhyme with *head*). Results were interpreted by Steacy et al. within a statistical learning framework (i.e., a child-by-word framework), supporting a developmental model of word reading in which children are more likely to use an alternative vowel pronunciation in a nonword as they become more proficient readers and as the occurrence of the alternative vowel pronunciation is increasingly supported by the corpus of words experienced. As would be expected in such a model of word reading development, child and corpus attributes work to tune variant vowel pronunciations across individual children and words, with important variance associated with both factors.

Difficulties Predicting Word Difficulty in Developing Readers

These relatively simple examples – varied vowel pronunciations of the digraph *ea* in single-syllable words (e.g., *-eat* pronunciations) and nonwords across two fairly frequent rime patterns (*-eam* vs. *-ead*) – illustrates one of the most persistent challenges posed by quasiregular orthographies such as English: accurately predicting word reading difficulty in developing

readers while accounting for word characteristics such as frequency, age of acquisition, phonological and morphological complexity, orthographic neighborhood size, and concreteness. Predictive models are essential for quantifying how word-level features interact with child-level knowledge and skill across development – for example, capturing the shift from reliance on simple spelling–sound correspondences to greater use of morphological and syllabic structure during word reading development (see Castles, Rastle, & Nation, 2018) – while also enabling the testing and refinement of theoretical accounts of lexical development and word recognition. Such models provide a principled framework for examining the child-by-word interactions that underpin variability in reading performance, particularly in populations with reading difficulties, and for understanding how instructional components influence these interactions (e.g., Steacy et al., 2016). Ultimately, understanding these complex interactions between child knowledge – in this case, word reading skill – and word features is critical for developing instructional systems that address the needs of individual students as they learn to read (e.g., Austin et al., 2022; Papalexandratou et al., 2024; U.S. Department of Education, 2023).

Achieving this goal, however, is complicated by the quasi-regular nature of English, which poses significant challenges for generating accurate predictive models across a wide range of words. Existing models tend to be underspecified and often biased by our limited ability to fully capture the quasiregular relations that determine word reading difficulty across words and children (e.g., Lim et al., 2024; Siegelman et al., 2020; Steacy et al., 2022). As a result, the explanatory power of word difficulty predictive models is constrained, reducing their utility for advancing our theoretical understanding of word reading development and providing limited input to instructional design.

To illustrate the nature and potential severity of the problem of predicting difficulty across a broad range of word characteristics, we pooled child and word data across multiple studies with the intent of developing a word-level prediction model of accuracy that could be used for predicting items difficulty (Grade Equivalent) on the 63 words from the Woodcock-Johnson-III Letter-Word Identification (WJ3-LWID; Woodcock et al., 2001) task. Accuracy data from 349 words pooled across five datasets, along with 16 word-level predictors (representing psycholinguistic word properties as well as measures of behavioral performance from the adult ELP), were used as training data to develop a general word difficulty predictive model which was subsequently applied to predict item difficulty on the WJ3-LWID items (for details on the sample and analytic approach see the supplemental materials). Based on the pooled data an additive linear regression model predicting (known) accuracy scores was fit from all predictors with the adjusted- R^2 of this model being 71.7%. This model was applied to the list of 63 WJ3-LWID items (pseudo- R^2 was estimated at 60.4%) and the relation between the predicted accuracy scores and WJ's GE scores plotted (Figure 2). The plot suggests that the predictive model of word difficulty is quite accurate for relatively simple words (i.e., lower right of figure) but loses precision as words become more complex (i.e., upper left), as indexed by the distance of the word from the regression line. These results underscore the need for a large-scale behavioral database capturing word reading difficulties across a wide range of word characteristics while sampling the full age range of reading skills in English-speaking children in the early stages of learning to read.

Figure 2 Here

In sum, ample evidence indicates that the quasiregular nature of the English orthography places significant and unique pressures on the development of children's word reading skills,

resulting in pronounced child-by-word effects over the course of development. Understanding the complex interplay between children's skills and knowledge and the characteristics of words across development remains a critical challenge from both theoretical and instructional perspectives. To help address this need, we introduce the d-ELP – a companion to the English Lexicon Project (Balota et al., 2007) – as an initial resource for examining how English orthography shapes word reading development in children across a large corpus of words. This database provides continuous IRT-based word reading difficulty estimates for 9,961 of the most frequently printed English words for U.S. children in 1st to 5th grade, along with detailed word-level characteristics, and raw trial-level data. In what follows, we detail the methods used to collect the data; analytic framework used to establish a continuous measure of word reading difficulty; descriptive statistics of the database; an explanation of the functionality of the publicly available relational database that provides searchable access to word difficulty ratings and associated word-level statistics for each of the stimuli; and finally a demonstration of the d-ELP's ability to estimate the impact of general word-level predictors (e.g., frequency, word length, transparency, semantic measures, and orthographic neighborhood density) on word recognition performance in developing readers.

Method

Design and Procedures

Data collection procedures varied across the project timeline and are described in three formats (Waves 1-3) for the primary data collection, with supplemental archival data included from a prior study. Changes to the data collection procedures across Waves 1-3 were made to increase the probability that children would complete their assigned list of words while also trying to maximize the number of responses per word across the sample. During all aspects of

data collection, children worked one-on-one with a trained research assistant in their school, read printed words aloud, and had each response scored as correct or incorrect without performance feedback. For Waves 1-3, children completed only the word reading tasks, with sessions lasting approximately 20 minutes. In contrast, archival participants completed word reading as part of a larger assessment battery. Two sets of linking items, comprising 99 total words, were used throughout Waves 1-3; the 63 words from the discontinued WJ3-LWID and 36 words from the NIH Toolbox (LaForte et al., 2024). For the supplemental archival data only the 63 WJ3-LWID items were available as linking items. Ceiling rules were only used on the WJ3-LWID, as described by the assessment manual, whereas all other items were only scored if the child attempted that item. Across all data collections, children provided verbal assent at the start of the session, and parental consent was obtained beforehand. Data collection procedures were approved by the University Institutional Review Board.

Waves 1-3

Children in grades 1-5, 1388 in total, were presented with a list of printed words (paper or tablet) and asked to read those words aloud. Scores reflected untimed accuracy on each word with words read correctly scored as a 1 and words read incorrectly scored as a 0. Self-corrections were counted as correct. The assessment ended immediately if a child became visibly frustrated during the word reading task or asked to stop.

Wave 1. In Wave 1, children were assigned to one of 200 preassembled word lists (40 lists per reading levels 1–5) designed to sample broadly from words at, above, and below their reading level. Before list assignment, children completed the WJ3-LWID task, beginning with the first item and following ceiling rules for stopping and scoring. Raw score (RS) performance on this task determined the grade-level list to which each child was assigned. The following

WJ3-LWID RS performance levels were used to assign children to reading level lists: less than 30 RS – Grade 1 list; 30 to 40 RS – Grade 2 list; 41 to 48 RS – Grade 3 list; 49 to 53 RS – Grade 4; and 54 and greater RS – Grade 5 list. The first set of children ($n=67$) in Wave 1 received a paper version of the word reading assessment in which the words were printed in columns (3 columns of 25 words per page) on physical sheets of paper. Later in Wave 1, students were presented with the same lists in the same order and same number of words and columns per page (again 3 columns of 25 words per page) but presented on an iPad screen rather than a physical sheet of paper. Students still saw words in a list with columns printed in black words on a white background on an iPad screen. We had hoped to establish ceiling rules for lists that would allow us to assign all words above the ceiling as attempted and incorrect. However, there was enough within- and between-child variability on list reading that ceiling rules were deemed impossible, leading us to only score those items that the child attempted.

Wave 2. In Wave 1 we encountered issues of children not finishing their assigned list for various reasons (e.g., inadequate allotment of assessment time or frustration/stamina issues), resulting in some children not attempting all embedded NIH toolbox linking items. Thus, in Wave 2, all 99 linking items (63 WJ3-LWID and 36 NIH Toolbox) were given first so that all children would attempt all linking items. Additionally, in Wave 1 the probability that a word would be attempted by a child decreased the further it was on the list (i.e., it was related by how far a child got on the list), and therefore in Wave 2 the order of words was randomized.

Wave 3. In Wave 3 we worked to better match words difficulty to a child's ability level using initial IRT model results. This was done by first converting their WJ3-LWID performance to an estimated theta based on a preliminary Rasch model. Words with the highest standard error (highest standard error items were chosen because it suggested more data is needed for that item)

within a range around the estimated theta were selected for reading lists to increase the number of responses. The range in which words were selected around the estimated theta changed several times during collection of Wave 3 data collection based on feedback from testers to increase the number of responses from children. The number of pages varied by level with most having 400 words selected but at the lower end fewer words were selected so students would be attempting only the most relevant words, with words randomly ordered and split across lists of 50 words. Across Waves 1-3 the maximum number of pages to be attempted was 8.

Additional Archive Data

Data from 519 children who attempted 263 of the same words in a prior data collection were added to the d-ELP sample to increase the overall sample size as well as increase the number of attempts for these select words. These children also completed the WJ3-LWID which served as linking items for this sample. This sample was presented with words in a traditional paper format. Children from this data collection have student ids greater than 99000 in the d-ELP database.

Participants

Our sampling philosophy for participants was to assess the full range of reading skills across children from the selected grade ranges and allow every child to contribute to the database to the fullest of their ability. Thus, we did not have strict inclusion/exclusion criteria for participation; however, we did ask teachers to identify children within their classroom for which this study was not appropriate, and these children were excluded from assent and assessment. Data was collected across schools, both public and private, in five different U.S. States (Connecticut, Florida, Georgia, Maryland, and Pennsylvania). Participants consisted of 1907

children in grades 1-5¹; with 331 – 1st graders, 474 – 2nd graders, 502 – 3rd graders, 360 – 4th graders, 217 – 5th graders, and 18 with unknown grade level. Of the 1708 students with completed teacher surveys, 49.6% were female; 36.8% White, 29.8% African American, 14.9% Hispanic, 3.0% Asian, 0.3% Native Hawaiian/Pacific Islander, 0.1% American Indian/Alaskan Native, and 4.6% Multiracial; 5% spoke African American English, 8% were English Language Learners, and 30% had a disability classification of some type, with the largest percentage having reading difficulties. Standardized WJ3-LWID performance across the sample of participants was $M=101.68$, $SD=15.03$, RS min=39, and RS max=149.

Stimuli

Our initial corpus consisted of approximately the 15,000 most frequently printed words in English for school age children from The Educator's Word Frequency Guide (Zeno et al., 2005). We trimmed the list of words in the following ways: 1) removed all hyphenated words, words with spaces, and words with apostrophes; 2) removed all first names; 3) retained names of major cities, states, and countries; 4) retained names of historical figures (e.g., Einstein); 5) retained names of groups of people (e.g., Egyptians); 6) removed approximately 1700 simple plurals – addition of s with no shifts in orthography or phonology (e.g., bats) – to increase the number of unique words in the database; 7) retained plurals that change orthography or phonology (e.g., cities, changes); 8) added back a set of 75 simple plurals (roughly matched to 75 irregular plurals) to allow for a plurals analysis, and 9) removed words that might be considered objectionable to children, teachers, or parents. This resulted in a list of 10,056 to which we added the linking items. We applied the difficulty prediction model/routine described above and rank ordered the words from easiest to most difficult. We selected the first 10,000 words from this list

¹ 5 students were in grade 6 from the archive data but reading below grade level.

(i.e., eliminating 56 words), retaining all linking items, representing the set of words to sample reading accuracy in children grades 1-5.

Graded word lists were initially constructed for Wave 1 using the rank order list with the NIH linking items mixed throughout the list in position of expected difficulty and the WJ3-LWID removed (these items were administered to identify the grade level list appropriate for each child). Forty lists were built at each reading grade level (1-5) with the idea that lists would sample from a wide range of difficulties while oversampling at the grade level of the list. We anticipated that children at a 1st grade reading level would be able to read fewer words during an assessment session and so 1st grade lists were 400 words, whereas 2nd – 5th grade lists contained 500 words. The Grade Equivalents (GE) of the 63 WJ3-LWID items were used as anchors within the 10,000 words to gauge the general difficulty of words for over sampling at a particular reading grade level.

Sampling for each list varied by reading grade level. Specifically, 1st grade lists systematically sampled 200 words rank ordered from 1→1000 difficulty, 100 from the 1001→2000 difficulty, 50 from the 2001 → 3000 difficulty, and 50 from 3001→5000 difficulty; 2nd grade lists systematically sampled 100 words from 1→2000 difficulty, 300 from the 2001→5000 difficulty, 50 from the 5001→6000 difficulty, and 50 from 6001→8000 difficulty; 3rd grade lists systematically sampled 100 words from 1→3000 difficulty, 300 from the 3001→6000 difficulty, 50 from the 6001→7000 difficulty, and 50 from 7001→9000 difficulty; 4th grade lists systematically sampled 100 words from 1→4000 difficulty, 300 from the 4001→7000 difficulty, 50 from the 7001→8000 difficulty, and 50 from 8001→10000 difficulty; and 5th grade lists systematically sampled 100 words from 1→4000 difficulty, 100 from the 4001→6000 difficulty, 200 from the 6001→8000 difficulty, and 100 from 8001→10000

difficulty. Lists within each grade level can be considered roughly equivalent in terms of overall word difficulty with lists becoming systematically more difficult across grade levels.

Apparatus

A majority of the students completed the word reading assessment on an iPad (whereas the first 67 administrations were presented in a typical paper list format). The iPads were 9th generation iPads with 10.2in retina displays with a resolution of 2160 x 1620 pixels at 264 pixels per inch. The iPad administration was powered using a Shiny app (Chang et al., 2020) created in R and deployed on the web via shinyapps.io. The administration included a two-screen solution such that the child had an iPad and the tester had an iPad each with a unique but linked display. The tester would first enter the student id into both the student and tester iPad. This would either bring up the appropriate list for that child (when using preassigned fixed form lists) during Wave 1 or set the seed for word randomization to be the same on both the student and tester versions during Wave 2-3 (for later versions with randomization to ensure words were randomized yet in the same random order on both the student and tester iPads). The student iPad presented words in black 30px font on a white background in 3 columns of 25 words. The tester iPad looked similar but had a checkbox for scoring next to each word. Testers would select the words which the child read incorrectly (allowing for unchecking for self-corrections, and data were only sent to the database once the tester hit submit). In Waves 2 and 3, when a child did not complete the full list, on the page that a child stopped, the tester would click a button that said “stopped on this page” and then select the last item that the child attempted; the rest of the words (which were not attempted) would be entered into the database as missing. (Wave 1 data underwent a manual data correction procedure to account for where kids stopped before this feature was added to the app.) After completion of a page, both the student and the tester would hit the “Next” button at the

bottom right of their screen, which would present the next list to be read. After the assessment was complete, the tester hit the “submit” button to send the item response data via the web to secure database storage.

Analytic Approach

Creating Difficulty Estimates

Before moving to IRT modeling, items were explored using classical test theory in order to evaluate which items were appropriate to include in the IRT model. Because of the sampling procedures of items to children, based on expected item difficulty and child ability, proportion correct was not included in this evaluation. Number of correct and incorrect responses along with item correlations with WJ3-LWID total score were examined. A total of 67 items were removed from the subsequent IRT analysis for having negative correlations with the latent word difficulty trait or not having at least 2 correct and incorrect responses, suggesting the items estimates were not trustworthy. These were removed because items with these characteristics may lead to misleading estimates but do not necessarily say anything about the item itself given our sampling procedures (i.e., it may just be due to insufficient data for that item). This resulted in a total of 9,961 items for model fitting and difficulty estimation.

Next, a Rasch model (with planned missing design) was fit to the child-level data using the TAM package (Robitzsch et al., 2024) in R using marginal maximum likelihood estimation². The final dataset used here consisted of 631,322 total responses from 1907 children across 9,961 words. This package was selected because it offered the capability to model difficulty in the presence of missing data (i.e., not all children saw all words). However, it should be noted that these procedures assume that data are missing completely at random. Although it is true that data

² A Bayesian-based Rasch model was also fit to the data with the correlation between estimates using traditional and Bayesian approaches exceeding .97. We therefore report results from the traditional Rasch approach.

were missing because the child did not attempt that word, which words were attempted and not attempted did relate in some way to the child's ability level in at least 2 ways: 1) children were assigned to lists of words to attempt based on likely ability level and 2) during Wave 1 words were ordered by potential difficulty with children attempting easier words first, children often ran out of time/energy before completing the list and tended to attempt less words above their ability level. It's possible that there is some relation between whether an item was attempted and a child's ability level (i.e., difficult words were less likely to be attempted by poor readers). However, given the linking items attempted by all children span the full range of the item difficulty distribution, we believe this anchored the analysis in a way that allowed for item difficulty to be estimated while taking the ability of the students who attempted that item into account. Because the number and reading skill of students who attempted each item differed, it's not the case that estimates for items with more attempts are necessarily more precise because the precision of the estimate also depends on whether those students represented an appropriate level of ability for that item. For this reason, we avoid using number of attempts in the discussion of precision of individual estimates but still present this data in the relational database. The dataset was collected using a planned-missing design, thus traditional residual-based global fit indices were not computable. Specifically, the standardized root mean square residuals (SRMR, SRMSR) and Q3-based statistics returned infinite or undefined values, which is expected when item pairs share limited overlap or when some items exhibit near-zero variance among respondents. Despite this, alternative global fit indices indicated acceptable overall model performance. The mean absolute deviation of item covariances ($100 \times \text{MADCOV} = 3.46$) suggested that the Rasch model adequately reproduced the observed covariance structure among items. The Global Homogeneity Parameter ($\text{GHP} = 0.33$) was lower than the ideal value of 1.0,

indicating that the model underestimates the variability present in the observed data. This underfit likely reflects the sparse response matrix and not substantive model misspecification. Taken together, the pattern of results suggests that the Rasch model provided a reasonable representation of the data within the constraints of the planned-missing design, though localized dependencies or multidimensionality cannot be ruled out.

Most of the items did not have what is typically considered to be a sufficient sample size for this type of model. The average number of responses per word for nonlinking items was 47.14 with a range of 10-592 responses/word. Given this variability and its effects on difficulty rating, we provide standard errors associated with the difficulty estimate to help users gauge the certainty associated with a given difficulty estimate. It is important to interpret the difficulty estimates in the context of the standard error to note the uncertainty surrounding that estimate. We also provide 95% confidence intervals for the difficulty estimates using this standard error to help the user see the plausible value range for that estimate.

Creating Grade Equivalent Scores

Grade equivalent scores were created by first fitting a function for the WJ3-LWID linking items between the IRT derived difficulty scores and the grade equivalent score from the WJ3-LWID subtest corresponding to the associated raw scores. Plotting the relation between the IRT derived difficulty and WJ grade equivalent scores revealed a very strong, albeit non-linear, relation (see Figure 3). We therefore fitted a function with linear, quadratic, and cubic terms to predict grade equivalence from IRT difficulty, which indeed accounted for 97% of variance in WJ items. The coefficients from this regression were then used to estimate grade equivalent scores from difficulty scores on the full sample of items. This variable is referred to as “grade_eq” in the database.

Figure 3 Here

Word Characteristics

The d-ELP interactive database groups word characteristics into categories – General Identifier, Word Difficulty, Occurrence Measures, Semantics, Orthography, Phonology, Orthography-to-Phonology (O2P) and Phonology-to-Orthography (P2O), and Orthography-to-Semantics (O2S). We present variables in their current subgroups here to match those in the database. Wherever possible, we retained the names of variables currently available in other databases (e.g., ELP; Balota et al., 2007; SCOPE; Gao et al., 2023). Below are definitions of the current variables in the database, however we anticipate additional variables will be included in the d-ELP database in the future. The description of each variable can be found in the d-ELP variable dictionary along with the citation for that variable. As new variables are added, the associated descriptions and citations will be added to the variable dictionary on the website as well.

General Identifier

Word_Lowercase. The lowercase version of the word, helpful for merging data with varying capitalizations.

Word. Depicts the word (with associated capitalization) as it was presented to the students in the d-ELP word reading assessment.

Word Difficulty

Difficulty. Word reading accuracy item difficulty estimate derived from the Rasch model estimation with lower values representing easier items and higher values representing more difficult items.

Standard_Error. Standard error associated with the item difficulty derived from the Rasch model.

Difficulty_Lower. Lower bound of the 95% confidence interval around the difficulty estimate. This value was calculated as $\text{difficulty} - 1.96 \times \text{standard_error}$.

Difficulty_Upper. Upper bound of the 95% confidence interval around the difficulty estimate. This value was calculated as $\text{difficulty} + 1.96 \times \text{standard_error}$.

Grade_Eq. Grade equivalent of the item difficulty based on the WJ3-LWID grade-based norms. More information on how this was calculated can be found in the “creating grade equivalent scores” section above.

N_Attempts. Number of children that attempted to read the word.

N_Correct. Number of children that correctly read the word.

N_Incorrect. Number of children that incorrectly read the word.

Occurrence Measures

General Frequency

Tasa_SFI. Standard Frequency Index taken from Zeno et al. (1995) which is a logarithmic transformation of a word’s frequency in text.

Tasa_D. Dispersion value taken from Zeno et al. (1995) which represents the number of different source documents in which the word appeared.

Tasa_U. Use variable taken from Zeno et al. (1995) which refers to the number of grade level corpora in which the word occurred. This value ranges from 1 to 4 and indicates the number of grade ranges over which the word was found. The four grade bands were K-2, 3-4, 5-6, and 7-12.

Tasa_F. Frequency variable taken from Zeno et al. (1995) which represents the total raw frequency count of the word across the entire corpus. It indicates how often the word appeared in the 17.25-million word sample.

Tasa_Gr1-Tasa_Gr13. Grade-level frequency count that indicates how often a word appeared in texts intended for that U.S. school grade level Zeno et al. (1995).

Log_Freq_HAL. Log10 version of frequency norms based on the Hyperspace Analogue to Language (HAL) corpus (Lund & Burgess, 1996).

Freq_SUBTLEXUS. Log10 version of frequency norms based on the SUBTLEXUS corpus (Brysbaert & New, 2009).

Sub_Unigram_Freq. Frequency of a unigram in the complete subtitle archive of OpenSubtitles.org based on the values reported in subs2vec (van Paridon & Thompson, 2021).

Wiki_Unigram_Freq. Frequency of a unigram in Wikipedia text based on values reported in subs2vec (van Paridon & Thompson, 2021).

Cyplex_7_9_Raw_Freq. Number of times the word appears in books for the 7-9 age range of the Children and Young People's Books Lexicon (CYP-LEX) corpus (Korochkina et al., 2024).

Cyplex_7_9_Zipf_Freq. Standardized frequency metric (logarithmic scale) for words appearing in the 7-9 age range of CYP-LEX (Korochkina et al., 2024).

Cyplex_7_9_Book_Prop. Proportion of books in the 7-9 age range in the CYP-LEX corpus (Korochkina et al., 2024) in which the word appears.

Cyplex_10_12_Raw_Freq. Number of times the word appears in books for the 10-12 age range of the CYP-LEX corpus (Korochkina et al., 2024).

Cyplex_10_12_Zipf_Freq. Standardized frequency metric (logarithmic scale) for words appearing in the 10-12 age range of CYP-LEX (Korochkina et al., 2024).

Cyplex_10_12_Book_Prop. Proportion of books in the 10-12 age range in the CYP-LEX corpus (Korochkina et al., 2024) in which the word appears.

Contextual Frequency

CD_SUBTLEXUS. Log10 version of the contextual diversity of a word which refers to the number of passages (documents) in the SUBTLEXUS corpus (Brysbaert & New, 2009) containing a particular word.

Semantics

Familiarity

Prevalence_Brys. Proportion of people who know the word in Brysbaert et al. (2019) in which participants were asked to indicate whether they know a word in a list of words and nonwords in an online crowdsourcing study. Values were translated to z-values based on a cumulative normal distribution. A word known by 2.5% of participants corresponds to a prevalence of -1.96 and a word known by 97.5% of participants corresponds to a prevalence of +1.96.

Fam_Glasgow. A word's subjective familiarity on a 1 (unfamiliar) to 7 (familiar) scale (Scott et al., 2019).

Age of Acquisition

AoA_Kuper. Ratings in which participants were asked to enter the age at which they thought they had learned the word (Kuperman et al., 2012).

AoA_LWV. Age at which people acquired a word based on a three-choice test administered to participants in grade 4 to 16 (Brysbaert & Biemiller, 2017).

Part of Speech

DPoS_Brys. Dominant grammatical category to which a word is assigned in accordance with its syntactic functions (Brysbaert et al., 2012).

POS³. Word's part of speech based on the English Lexicon Project (ELP, Balota et al., 2007).

Concreteness

Conc_Brys. Represents word concreteness, the degree to which the concept can be experienced directly through the senses from a 1 (abstract) to 5 (concrete) scale (Brysbaert et al., 2014).

Polysemy

Nmeanings_Websters. Number of meanings based on the Websters dictionary (Gao et al., 2023).

Affect

Valence_Warr. Pleasantness of a word on a 1 (happy) to 9 (unhappy) scale (Warriner et al., 2013).

Valence_NRC. Word-emotion associations ranging from 0 (negative) to 1 (positive) scale (Mohammad & Turney, 2013).

Arousal_Warr. Intensity of emotion provoked by a word on a 1 (aroused) to 9 (calm) scale (Warriner et al., 2013).

Arousal_NRC. Passiveness or activeness of a word, 0 (low arousal) to 1 (high arousal), calculated via manual annotators who were presented with four words and ask to rank them by activeness and passiveness (Mohammad, 2018).

³ 53 words not available in ELP which were coded separately based on Oxford English Dictionary.

Emot_Assoc_Positive. Extent to which a word was associated with a positive emotion ranging on 0 (not associated) to 1 (associated) scale (Mohammad & Turney, 2013).

Emot_Assoc_Joy. Extent to which a word was associated with joy ranging on 0 (not associated) to 1 (associated) scale (Mohammad & Turney, 2013).

Semantic Diversity

Sem_Diversity. Degree to which different contexts associated with a word vary in their meaning (Hoffman et al., 2013).

Sem_N. Number of semantic neighbors based on co-occurrence (Shaoul & Westbury, 2006).

Morphology⁴

NMorph. Number of morphemes in a word based on the values in the English Lexicon Project (ELP, Balota et al., 2007).

MorphSp. Morphological spelling of a word based on the values in the English Lexicon Project (Balota et al., 2007).

Orthography

Number of Letters

Length. Number of letters in a word.

Orthographic Neighborhood

Ortho_N. Orthographic neighborhood size of a word representing the number of words that can be obtained by changing one letter while preserving the identity and position of the other letters (Balota et al., 2007).

⁴ 53 words not found in the ELP and were coded by a trained speech-language pathologist.

OLD20. Mean orthographic Levenshtein distance of the 20 closest orthographic neighbors of a word (Balota et al., 2007).

OLD20F. Mean Log_Freq_HAL of the closest 20 orthographic neighbors (Balota et al., 2007).

Orth_Spread. Number of letter positions that can be changed to form another word that differs by a single letter (Chee et al., 2020).

Bigram Frequency

BG_Sum. Summed bigram (two letter sequence) frequency taken from the English Lexicon Project (Balota et al., 2007).

BG_Mean. Average bigram frequency which is the summed bigram frequency divided by the number of bigrams in a word (Balota et al., 2007).

BG_Frequency_By_Position. Summed bigram frequencies that are sensitive to the positions within words (Balota et al., 2007).

Phonology

Pronunciation

Pronunciation. Machine-readable phonetic alphabet transcribed pronunciation (SAMPA; Wells, 1997) taken from the ELP (Balota et al., 2007)⁵.

Number of Phonemes

NPhon. Number of phonemes in a word based on the values in the ELP (Balota et al., 2007).

Number of Syllables

⁵ 53 words were coded by a trained speech-language pathologist.

NSyll. Number of syllables in a word based on the values in the English Lexicon Project (ELP, Balota et al., 2007).

Phonological Neighborhood

Phono_N. Number of phonological neighbors or the number of words that can be obtained by changing one phoneme in a word while preserving the identity and position of the other phonemes (Balota et al., 2007).

PLD20. Mean phonological Levenshtein distance of the 20 closest phonological neighbors of a word (Balota et al., 2007).

PLD20F. Mean Log_Freq_HAL of the closest 20 phonological neighbors (Balota et al., 2007).

Phonetic Features of the First Phoneme

First_Phon_Bilabial. Dichotomous variable of whether the first phoneme of a word is a bilabial (Gao et al., 2023; Weide, 2005).

First_Phon_Labiodental. Dichotomous variable of whether the first phoneme of a word is a labiodental (Gao et al., 2023; Weide, 2005).

First_Phon_Interdental. Dichotomous variable of whether the first phoneme of a word is an interdental (Gao et al., 2023; Weide, 2005).

First_Phon_Alveolar. Dichotomous variable of whether the first phoneme of a word is an alveolar (Gao et al., 2023; Weide, 2005).

First_Phon_Palatal. Dichotomous variable of whether the first phoneme of a word is a palatal (Gao et al., 2023; Weide, 2005).

First_Phon_Velar. Dichotomous variable of whether the first phoneme of a word is a velar. (Gao et al., 2023; Weide, 2005).

First_Phon_Glottal. Dichotomous variable of whether the first phoneme of a word is a glottal. (Gao et al., 2023; Weide, 2005).

First_Phon_Stop. Dichotomous variable of whether the first phoneme of a word is a stop (Gao et al., 2023; Weide, 2005).

First_Phon_Fricative. Dichotomous variable of whether the first phoneme of a word is a fricative (Gao et al., 2023; Weide, 2005).

First_Phon_Affricate. Dichotomous variable of whether the first phoneme of a word is an affricate (Gao et al., 2023; Weide, 2005).

First_Phon_Nasal. Dichotomous variable of whether the first phoneme of a word is a nasal (Gao et al., 2023; Weide, 2005).

First_Phon_Liquid. Dichotomous variable of whether the first phoneme of a word is a liquid (Gao et al., 2023; Weide, 2005).

First_Phon_Glide. Dichotomous variable of whether the first phoneme of a word is a glide (Gao et al., 2023; Weide, 2005).

First_Phon_Voiced. Dichotomous variable of whether the first phoneme of a word is voiced (Gao et al., 2023; Weide, 2005).

Orthography-to-Phonology (O2P) and Phonology-to-Orthography (P2O)

Phonographic Neighborhood

Phonographic_N. Number of phonographic neighbors which refers to the number of words that can be obtained by changing one letter and one phoneme while preserving the identity and position of the other letters and phonemes (Balota et al., 2007).

Phonographic_N_Freq. Average Log_Freq_HAL of the phonographic neighborhood of a word.

Spelling-to-Pronunciation Transparency Ratings

*Spell2Pron*⁶. Spelling-to-pronunciation transparency ratings which subjectively measure the ease of arriving at a correct pronunciation for a word by applying typical decoding rules, serving as a global index of regularity between orthography and phonology (Edwards et al., 2023).

Consistency

Consistency_Token_FF_O. Spelling-to-sound consistency measure of the word onset calculated as the log frequencies of friends divided by the total log frequencies of friends and enemies (Chee et al., 2020).

Consistency_Token_FF_R. Spelling-to-sound consistency measure of the word rime calculated as the log frequencies of friends divided by the total log frequencies of friends and enemies (Chee et al., 2020).

Consistency_Token_FB_O. Sound-to-spelling consistency measure of the word onset calculated as the log frequencies of friends divided by the total log frequencies of friends and enemies (Chee et al., 2020).

Consistency_Token_FB_R. Sound-to-spelling consistency measure of the word rime calculated as the log frequencies of friends divided by the total log frequencies of friends and enemies (Chee et al., 2020).

Orthography-to-Semantics (O2S)

Orthographic-Semantic Consistency

⁶ 807 words on the d-ELP list did not have spelling to pronunciation transparency ratings available from Edwards et al. (2023); additional data on these words using the same procedures they used in the original study so that all words in the d-ELP could have ratings available. Thus, there are 807 words with spelling-to-pronunciation transparency ratings that have ratings available in d-ELP that were not included in the original set from Edwards et al. (2023).

Siegelman_OSC. Orthographic-semantic consistency represents the level of congruence in meaning between a word and its orthographic neighbors. This value is derived from the mean cosine similarity between the word and its orthographic neighbors in semantic space (Siegelman et al., 2022).

Marelli_OSC. Orthography-semantics consistency measures the degree of semantic relatedness between a word and all words that contain the same letter string as the word (e.g., corn-corner; Marelli & Amenta, 2018).

Descriptive statistics for the word-characteristic variables included in the d-ELP are shown in Table 1. The d-ELP database allows a list of words to be submitted to generate characteristics for the list (see description of the Query function below) or generate a list of words based on a set of specific constraint (see description of the Filter command below). Within both functions the full set of word characteristics/variables can be accessed through drop down menus. The d-ELP also allows visualization of word sets through a three-dimension plotting routine (i.e., routine used to generate Figure 1 above) that allows rotations to better visualize relationships among word characteristics. The database also allows a complete download of the word database along with the full database of deidentified child-level item data (see the description of the Information and Downloads tab below).

Table 1 Here

Using the Interactive Database

Users can request access to the interactive database by visiting <https://d-elp.org/>. Prior to being granted access to the database, we ask users to register and agree to use the data for non-commercial purposes. Once access is granted, the interactive database provides multiple tools and pathways for retrieving and exploring the data. The most basic way is for users to download

the full word level database from the Information and Downloads tab. For users wanting to obtain word characteristics on a specific list of words, the Query tab provides the capability to input words and select which variables to obtain. Alternatively, the Filter tab can be used to find words that meet certain criteria based on its characteristics, filtering the list based on chosen variables of interest. Finally, the raw trail data can be directly downloaded via the from the Information and Downloads tab.

Query

The query tab allows users to input (by typing or copying and pasting from excel) a list of words (separated by spaces) for which they would like associated word characteristics, activated through numerous checkboxes, sorted into expandable accordion groups. Once the user hits submit, the words along with the selected variables will be presented in the table to the right which can be downloaded using the “Download as CSV” button at the top of the table. Words will appear in rows in the same order that they were entered and any words that do not appear in the database will still appear in the list but will return “NA” for all of the variables. An example is presented in Figure 4.

How to use Query Filter Visualize your words Information and Downloads About the Authors License Give us Feedback Recent Updates

Enter words (separated by a space):
apple banana wordnotindata

Download as CSV

word	difficulty	standard_error	grade_eq
apple	-3.11	0.46	1.90
banana	-2.29	0.55	2.06
wordnotindata	NA	NA	NA

General Identifier >

Word Difficulty >

☒ difficulty
☒ standard_error
☐ difficulty_lower
☐ difficulty_upper
☒ grade_eq

Occurrence Measures >

Semantics >

Orthography >

Phonology >

O2P and P2O >

O2S >

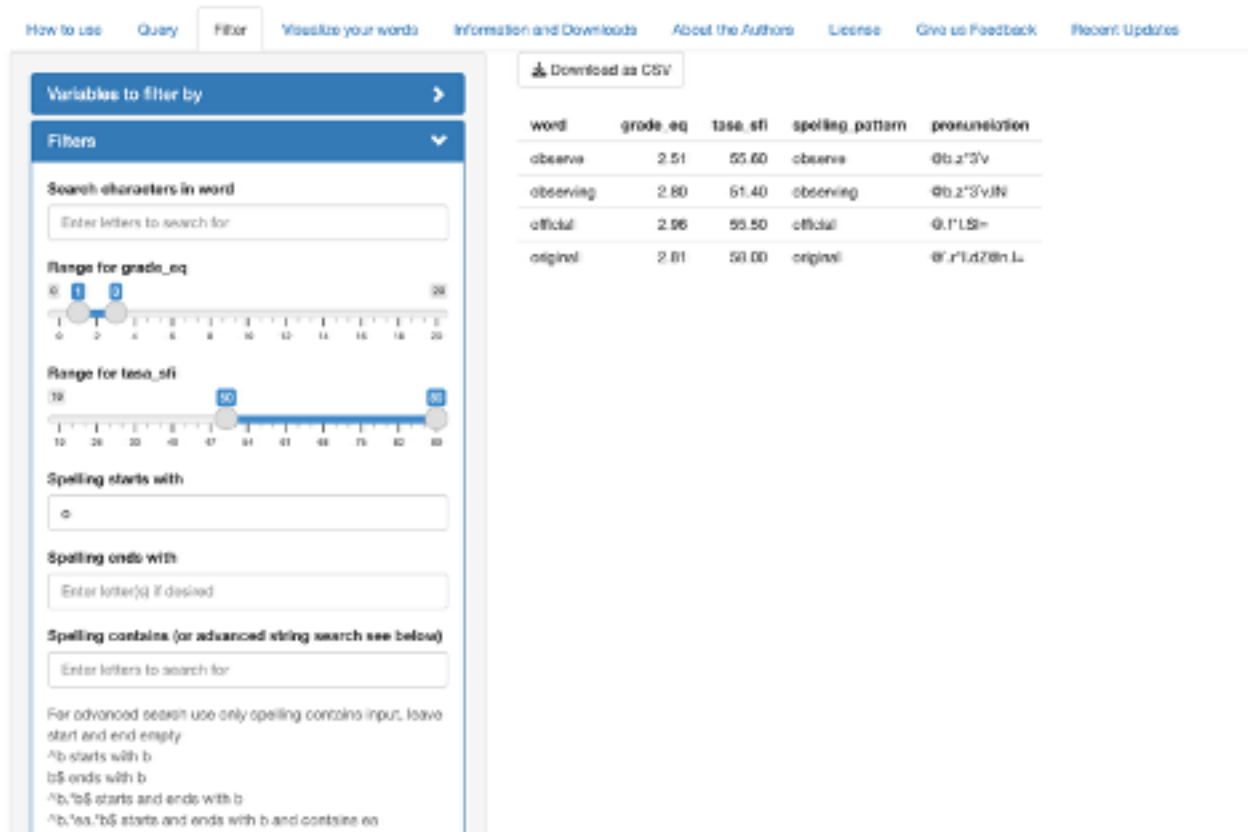
Submit

Figure 4

An example of the Query function of the searchable database.

Filter

The filter tab is used for finding words in the d-ELP list that meet certain criteria set by the user. First, users select the set of variables that they want to filter. It is important to note that if a word has missing data for any of the selected variables, then it will be removed from the list. Users can then provide specific criteria that words must meet on each of those variables. Numeric variables populate a range in which the upper and lower bounds can be moved to the desired number. Character string variables will populate text boxes to search for characters within that string. Figure 5 shows an example of a user looking for words with difficulties between grades 1 and 3, with standard frequencies above 50, starting with the letter o, and whose pronunciation starts with a schwa.

**Figure 5**

An example of the Filter function of the searchable database.

Similarly, Figure 6 shows a user searching for words with difficulties in the range of grades 2-4, frequencies above 50, and with the “ive” suffix in its morphological spelling. Note that this search is different than the spelling search above which would be searching for the letters “ive” in the spelling of the word. Here, since we are specifically looking at the morphological spelling it also returns words like “activity” which does not contain the letter string “ive” but does contain the morpheme.

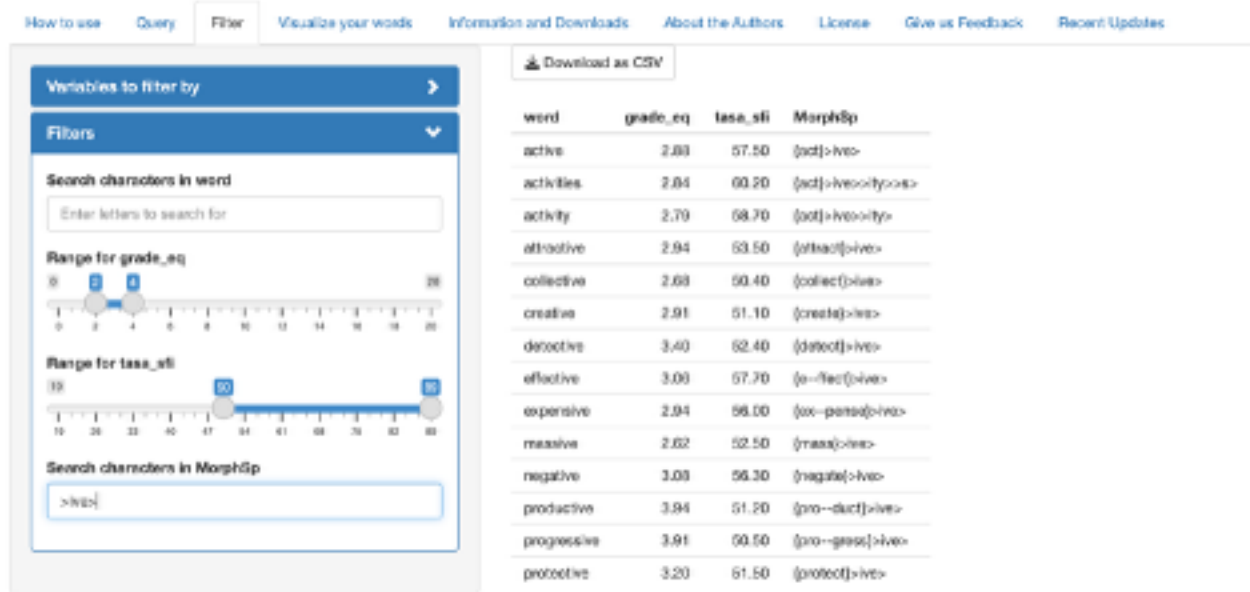


Figure 6

An example of the Filter function of the searchable database using the morphological spelling option.

Results

Interpreting Difficulty Scores

Difficulty estimates were derived from the IRT item difficulty parameter in which easier items have lower values and harder items have larger values. Difficulty had a mean of 0.41 and a standard deviation of 2.03, ranging from -7.02 to 9.15 (see Figure 7). It is important to interpret the difficulty estimates in the context of the associated standard errors to appreciate the uncertainty surrounding individual word difficulty estimates (i.e., the precision of the estimates). It is noteworthy that our average number of responses (47.14 attempts/item for nonlinking items) is in line with other “Lexicon Projects” across languages (for a summary of average observations per word across studies see Table 1, Stein et al., 2024), with the range being 20-55 average observations per word with one outlier being the Spanish Lexical Project (SPALEX; Aguasvivas et al., 2018) which had an average of 333 response per word. An important caveat is that the d-

ELP database presented here is the only “Lexical Project” to fit an IRT-based difficulty model across a sample of participants differing widely in word reading ability.

Figure 7 Here

As a tangible example of the interpretability of individual difficulty across items we compare two orthographically related and similarly difficult items “abandon” and “abandoned”. The item “abandon” has a difficulty estimate of 1.02 and “abandoned” has 1.07. While it is reasonable to interpret the difficulty of these words as being close to one another, it is incorrect to conclude that “abandon” is easier than “abandoned”. Looking at the confidence interval, the difficulty of “abandon” is likely to occur anywhere between 0.18 and 1.85 and the difficulty of “abandoned” could be anywhere between 0.08 and 2.06. With these overlapping ranges, we are unable to say for sure that the true difficulty of “abandon” is easier than “abandoned”; however, it would be appropriate to conclude that both items are easier than “abdomen” which has a difficulty estimate of 4.48 with a plausible value range between 3.32 and 5.63. And so, while caution is warranted when directly comparing difficulty estimates that are close in value; it is also important to note that the Rasch model empirically orders item difficulties by drawing on the full information available in the data and thus order does matter.

Interpreting Grade Equivalent

Grade equivalent (GE) scores are the difficulty scores described above translated to be on the WJ3-LWID grade equivalent scale norms. These norms were published in 2001 and may not reflect the true performance of students in that grade today. Estimates of difficulty derived from students in grades 1-5 were converted based on WJ3-LWID norms data with words receiving GE scores outside of the grade range of the sample (i.e., 0.4 to 19.7 GE). Similar to difficulty scores,

GE scores contain a range of uncertainty around the estimate; words with larger standard errors will have more uncertainty in the grade equivalent scores.

Example: Identifying Key Control Variables

There are numerous uses for this database; here we present one example of how data generated from the d-ELP database can be used to explore the impact of word characteristics (e.g., frequency, word length, transparency, semantic measures, and orthographic neighborhood size) as predictors of word difficulty to facilitate recommendations of the control variables to be included in factorial experiments and individual differences studies with developing readers. In this set of regression analyses we use the following variables as estimates for potentially important control variables representing each of the following word traits: frequency (Tasa_sfi), word length (Length), transparency (Spell2pron), semantic measures (Conc_Brys and AoA_Kuper), and neighborhood size (Ortho_N). We include two semantic measures as a means of highlighting the influence of word semantic characteristics on word reading difficulties in developing readers (see Kellenberger et al., 2024), variables that are often overlooked as control variables in developmental word reading studies. We demonstrate the overall effects of the potential control variables, as well as the unique R^2 contribution of each variable after controlling for the effects of the other 5 variables. Complete data from 6397 words are included in the analyses due to missing data across the various variables. Spell2pron was log transformed to correct for skewness and non-normality in the relation with difficulty. Correlation coefficients and descriptive statistics are presented in Table 2 with all correlations among variables being significant ($p < .001$). Correlations with word reading difficulty ranged from large (AoA_Kuper, Spell2pron, and Length), moderate (Ortho_N and Tasa_sfi), to small (Conc_Brys). In the initial multiple regression analysis, each of the 6 word characteristics were simultaneously entered as

predictors of difficulty with all but concreteness contributing significantly to the prediction of word reading difficulty (see Table 3, Model 1). Results suggest that words with more letters; appearing less often; being less transparent; rated to be learned later; and having fewer orthographic neighbors were more difficult to read. This model explained 64.5% of the variance in word reading difficulty for this sample of 6397 words.

To gauge the unique contribution of each variable in predicting word reading difficulty six additional regression models were run (see Table 3, Models 2-7). This is in keeping with the approach advocated for by Balota et al. (2012) of assessing the extent to which a novel predictor accounts for unique variance in megastudies, after controlling for other correlated variables. In these models the unique R^2 associated with each of the variables (represented by ΔR^2) was estimated by block entering the other five variables followed by adding the variable of interest. Overall, ΔR^2 values ranged from .02% to 11.2% of the variance, with the magnitude of unique contributions rank ordered (from largest to smallest) as follows: age of acquisition, transparency, orthographic neighborhood size, frequency, length, and concreteness. While the results clearly suggest there is considerable shared variance among the 6 predictors (67.9% of the explained variance is shared) there is a sizeable amount of the variance that is unique to the various predictors (32.1% of the variance explained is unique). Notably, variables representing age of acquisition, transparency, and orthographic neighborhood size accounted for the largest portion of unique variance but are largely missing as control variables in developmental experiential and individual differences studies of word reading. This is one example of how the d-ELP can allow researchers to improve their work by providing an evidence-based way to decide which variables to include (here as control variables, but possibly as independent variable as well) in their experimental designs.

Discussion

The English orthography is best described as quasiregular, consisting of a complex interplay between morphological regularities, syllabic constraints, and numerous sub-lexical regularities in spelling–sound correspondences. For this reason, English presents a set of unique and significant challenges for children as they work to develop efficient and automatic word reading skills (see Seymour, 2013; Seymour et al., 2003), with individual differences in word reading development representing a complex interaction between the property of words and the skills and knowledge that children bring to the process of learning to read (e.g., Steacy et al., 2022). Teasing out these complex interactions has been difficult due to the complex task of characterizing “word difficulty” from a developmental word reading perspective. The inability to precisely quantify word difficulty empirically across the wide range of words encountered during reading development has also been related to the lack of a large-scale developmental word reading database. To help alleviate these issues we developed the d-ELP database of word reading performance in a diverse sample of US children. In this first paper, we have used the data to further derive “true” estimates of word reading difficulty across a wide swath of words and to identify several key control variables to be used in future word reading studies. As with previous “lexicon projects” we have made our representational database publicly accessible, along with downloadable raw trail data, through the use of an interactive Shiny app. We believe this initial iteration of the d-ELP, and any subsequent versions, will have important educational, psychological, and computational uses going forward that will improve our basic understanding of reading development and practices of improving reading outcomes for children.

Limitations

Placing the 9,961 most frequently printed English words on a continuous difficulty scale represents a major advancement in promoting the understanding and modeling of word reading difficulty across the early reading developmental span. However, this achievement is best viewed as a foundational step. Because it is impractical to ask children to attempt all 9,961 words, the resulting database necessarily has limitations for modeling child-by-word interactions across both a wide developmental range and a large corpus of words (for details, see Keuleers, Diependaele, & Brysbaert, 2010). Nonetheless, the d-ELP provides a powerful tool for selecting words that match specific developmental periods and linguistic characteristics, thereby facilitating more precise and targeted data collection in which children can reasonably attempt all items. We emphasize that this initial iteration of the d-ELP represents a first step toward a more comprehensive understanding of how word characteristics influence reading difficulty in developing readers. The precision of the current word difficulty estimates is constrained by the number of child-level observations per item, but we aim to establish a mechanism for researchers to contribute additional data to improve item calibration, similar to our integration of archival data in the present study. In this sense, the d-ELP serves as an initial, open-ended framework for estimating and refining word reading difficulty across development.

Literature Cited

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Table 1*Descriptive Data for Selective Words Characteristics in the d-ELP*

Word Characteristic	N	Mean	SD	Min	Max
General Identifier					
Word	9961				
Word Difficulty					
Difficulty	9961	0.41	2.02	-7.01	9.14
Standard_Error	9961	0.49	0.09	0.06	0.85
Difficulty_Lower	9961	-0.54	2.02	-7.61	8.26
Difficulty_Upper	9961	1.37	2.04	-6.64	10.01
Grade_Eq	9961	3.23	1.30	0.47	19.73
N_Attempts	9961	63.38	179.51	10.00	1970.00
N_Correct	9961	42.57	117.80	2.00	1874.00
N_Incorrect	9961	20.81	103.76	3.00	1902.00
Occurrence Measures					
Tasa_SFI	9958	50.47	6.35	19.2	88.38
Tasa_D	9958	0.71	0.13	0.00	1.00
Tasa_U	9958	80.40	955.54	0.00	68006
Tasa_F	9958	1393.23	11782.68	1.00	495371
Log_Freq_HAL	9910	8.49	1.70	1.37	16.96
Freq_SUBTLEXUS	9740	2.48	0.80	0.30	6.31
Cyplex 7-9 RF	9834	1021.07	10123.20	1.00	608232
Cyplex 10-12_RF	9930	1985.09	19837.75	1.00	1171206
CD_SUBTLEXUS	9740	2.31	.69	0.30	3.92
Semantics					
Prevalence_Brys	8029	2.30	0.23	0.00	3.00
Fam_Glasgow.	3200	5.53	0.72	2.00	7.00
AoA_Kuper	6437	7.77	2.28	2.00	17.00
AoA_LWV	6163	7.71	2.19	2.00	17.00
Conc_Brys	7989	3.26	1.05	1.07	5.00
Nmeaning_Websters	7215	3.09	2.43	1.00	28.00
Valence_Warr	5858	5.34	1.27	1.40	8.53
Valence_NRC	6967	0.55	0.22	0.01	1.00
Arousal_Warr	5858	4.13	0.91	1.60	7.74
Arousal_NRC	6967	0.48	0.18	0.07	0.97
Emot_Assoc_Pos	5825	0.19	0.39	0.00	1.00
Emot_Assoc_Joy	5825	0.06	0.24	0.00	1.00
Sem_Diversity	9508	1.69	0.31	0.30	2.41
Sem_N	9726	2888.79	3094.36	0.00	9931.00
NMorph	9961	1.78	0.72	1.00	5.00
Orthography					
Length	9961	6.94	2.19	1.00	16.00
Ortho_N	9910	3.39	5.02	0.00	34.00
OLD20	9782	2.35	0.82	1.00	7.20
OLD20F	9782	7.11	0.85	3.87	11.90

Ortho_Spread	9560	1.12	1.23	0.00	5.00
BG_Sum	9908	25553.46	14692.38	95.00	107720
BG_Mean	9908	4122.96	1604.26	95.00	14646
BG_Freq_By_Position	9908	3984.26	2110.07	2.00	13208
Phonology					
NPhon	9961	5.76	2.03	1.00	19.00
NSyll	9961	2.15	0.96	1.00	6.00
Phono_N	9910	7.40	11.31	0.00	60.00
PLD20	9782	2.26	0.95	1.00	7.90
PLD20F	9782	7.28	1.02	3.18	12.01
Ortho-to-Phono (O2P)					
Phonographic_N	9910	1.99	3.50	0.00	24.00
Phonographic_N_Freq	4729	6.79	2.10	0.00	15.00
Spell2Pron	9961	2.21	0.55	1.00	5.27
Consist_Token_FF_O	9561	0.87	0.15	0.00	1.00
Consist_Token_FF_R	9561	0.71	0.21	0.00	1.00
Consist_Token_FB_O	9561	0.83	0.17	0.00	1.00
Consist_Token_FB_R	9561	0.57	0.24	0.00	1.00
Ortho-to-Semantics (O2S)					
Siegelman_OSC	8014	0.15	0.20	-0.23	0.98
Marelli_OSC	5334	0.81	0.27	-0.09	1.00

Table 2*Descriptive Statistics and Correlation Coefficients for the Control Variable Analyses (N=6397)*

Variable	Mean	SD	Difficulty	Frequency	Length	Trans	Concrete	AoA	Ortho N
Difficulty	0.14	2.11	–						
Frequency	51.13	6.65	-.48	–					
Length	6.62	2.28	.53	-.29	–				
Log Trans	0.74	0.26	.53	-.18	.49	–			
Concrete	3.38	1.07	-.26	-.10	-.29	-.16	–		
AoA	7.75	2.26	.68	-.48	.42	.29	-.37	–	
Ortho N	3.61	5.51	-.49	.24	-.64	-.48	.24	-.35	–

Note. Trans = Spelling-to-pronunciation Transparency Ratings; Concrete = Concreteness Rating; AoA = Age of Acquisition; Ortho N = Orthographic Neighborhood Size.

All correlations are significant ($p < .001$).

Table 3

Hierarchical Regression Analyses Predicting Word Reading Difficulty using the Potential Control Variables (N=6397) along with Estimates of the Size of the Contribution of each Control Variable to Variance Explained (ΔR^2)

Variable	Combined Model				
	<u>Unstandardized Coefficients</u>				
	B	SE	β	<i>t</i>	<i>p</i>
Constant	-2.469	0.242		-10.195	<.001
Frequency	-0.053	0.003	-0.166	-18.237	<.001
Length	0.070	0.010	0.076	7.222	<.001
Log Trans	2.529	0.072	0.309	34.962	<.001
Concrete	-0.033	0.017	-0.017	-1.946	0.052
AoA	0.414	0.009	0.443	44.901	<.001
Ortho N	-0.037	0.004	-0.096	-9.509	<.001

Model 1 – Combined: $R^2 = .645$, $F(6, 6390) = 1934$, $p < .001$

Model 2 – Unique Frequency Effects: $\Delta R^2 = .019$, $F(1, 6390) = 332.57$, $p < .001$

Model 3 – Unique Length Effects: $\Delta R^2 = .003$, $F(1, 6390) = 52.154$, $p < .001$

Model 4 – Unique Transparency Effects: $\Delta R^2 = .068$, $F(1, 6390) = 1222.3$, $p < .001$

Model 5 – Unique Concreteness Effects: $\Delta R^2 = .0002$, $F(1, 6390) = 3.787$, $p = .052$

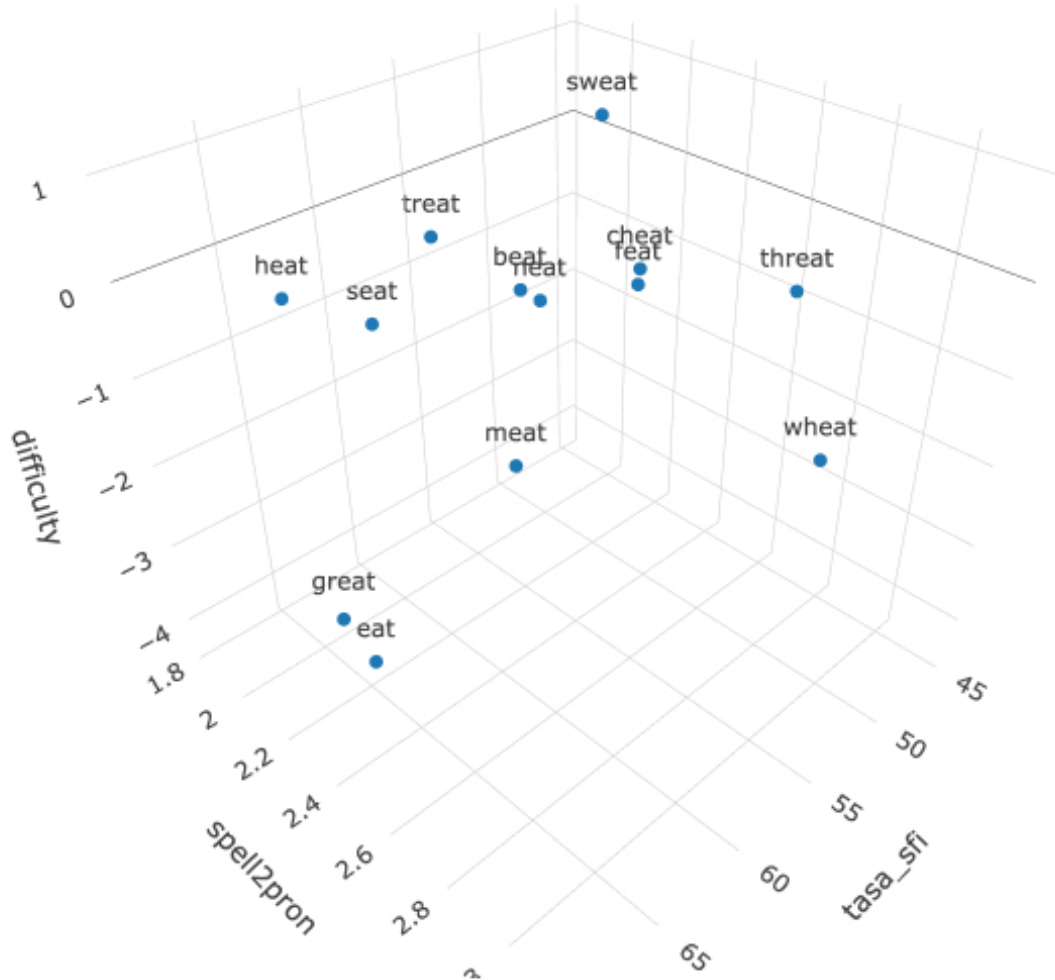
Model 6 – Unique Age of Acquisition Effects: $\Delta R^2 = .112$, $F(1, 6390) = 2016.1$, $p < .001$

Model 7 – Unique Ortho Neighborhood Effects: $\Delta R^2 = .005$, $F(1, 6390) = 90.426$, $p < .001$

Note. Model 1 represents the simultaneous addition of predictors into the prediction model; Models 2-7 represent the unique R^2 associated with each of the listed variables when added last to the model in a block entry fashion; Trans = Spelling-to-pronunciation Transparency Ratings; Concrete = Concreteness Rating; AoA = Age of Acquisition; Ortho N = Orthographic Neighborhood Size.

Figure 1

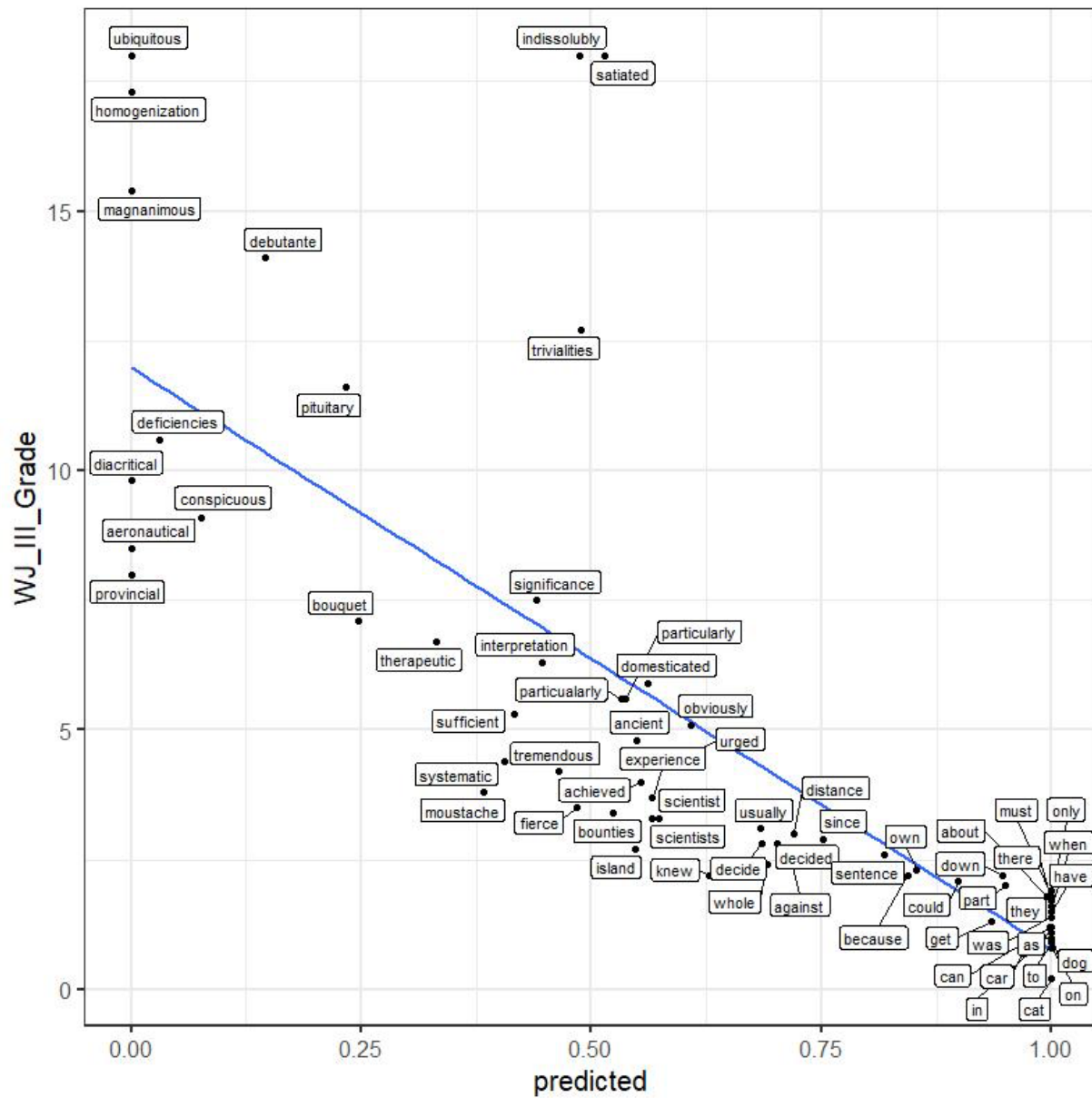
Word Difficulty (difficulty) as a Function of Frequency (tasa_sfi) and Regularity (spell2pron) for Single Syllable Words with the -eat Rime Pattern.



Note. tasa_sfi= Standard Frequency Index (Zeno et al., 1995); spell2pron = Spelling to Pronunciation Transparency (Edwards et al., 2024)

Figure 2

Grade Equivalent (GE) scores (y-axis) as a Function of Predicted Accuracy (x-axis) in WJ-LWID items. Blue trendline is best linear fit.



Note. Blue trendline is best linear fit.

Figure 3

Nonlinear Relationship Between WJ3-LWID Grade Equivalent Estimates and IRT Derived Difficulty for the 63 WJ3-LWID Items.

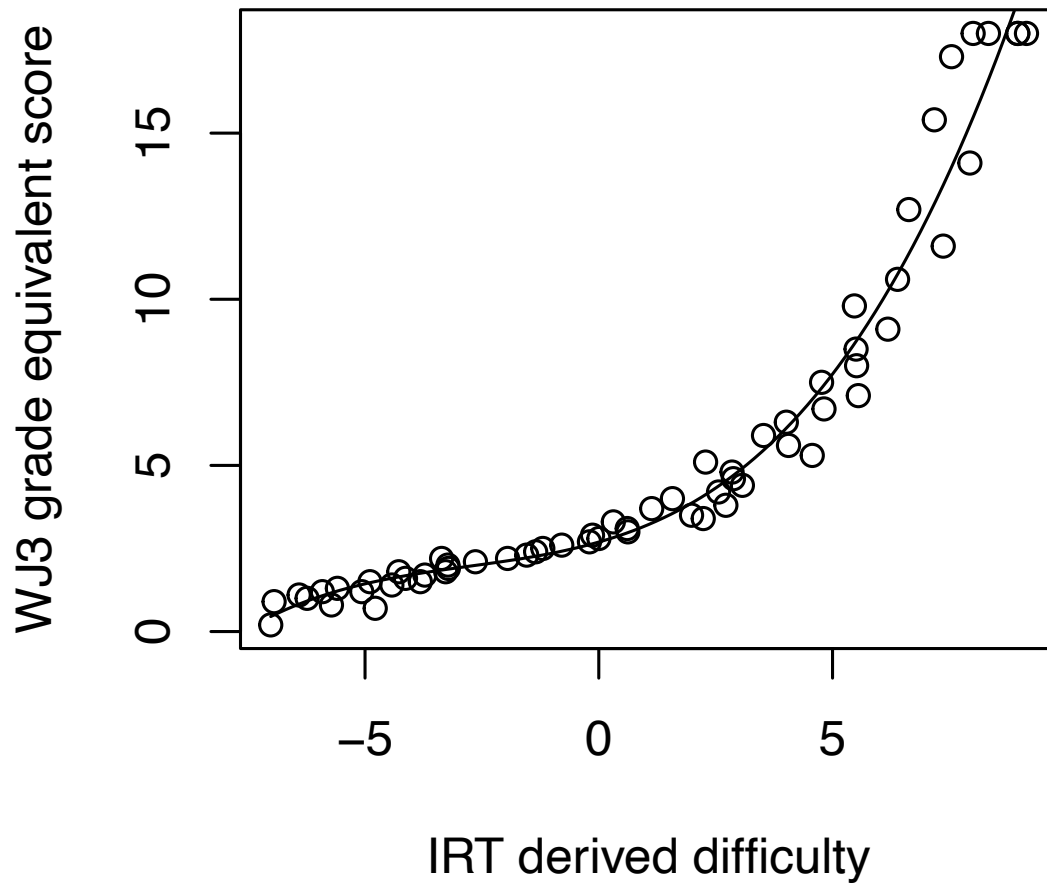


Figure 7

Histogram of difficulty estimates from the Rasch IRT model.

